

physically sensible crystal must have a unit cell with nonzero volume, hence, we require any lattice matrix to be non-singular.

Fractional coordinates express the location of an atom using the lattice vectors as the basis vectors. For instance, an atom with fractional coordinates $\mathbf{x} = (0.2, 0.3, 0.5)^\top$ has Cartesian coordinates $\tilde{\mathbf{x}} = 0.2\mathbf{l}^1 + 0.3\mathbf{l}^2 + 0.5\mathbf{l}^3$. The periodicity in fractional coordinates is defined by the (flat) unit hypertorus, i.e., we have the equivalence relation $\mathbf{x} \sim \mathbf{x} + \mathbf{k}, \mathbf{k} \in \mathbb{Z}^3$. We can convert between fractional coordinates $\mathbf{X}$ and Cartesian coordinates $\tilde{\mathbf{X}}$ as follows:

$$\tilde{\mathbf{X}} = \mathbf{L}\mathbf{X}, \quad (\text{A2})$$

$$\mathbf{X} = \mathbf{L}^{-1}\tilde{\mathbf{X}}. \quad (\text{A3})$$

A.2 Invariance and equivariance in periodic materials

The energy per atom $\epsilon(\mathbf{M}) = E(\mathbf{M})/n$ of a periodic material $\mathbf{M} = (\mathbf{X}, \mathbf{L}, \mathbf{A})$ has several invariances.

- Permutation invariance: $\epsilon(\mathbf{X}, \mathbf{L}, \mathbf{A}) = \epsilon(\mathbf{P}(\mathbf{X}), \mathbf{L}, \mathbf{P}(\mathbf{A}))$ for every permutation matrix $\mathbf{P}$.
- Translation invariance: $\epsilon(\mathbf{X}, \mathbf{L}, \mathbf{A}) = \epsilon(\mathbf{X} + \mathbf{t}, \mathbf{L}, \mathbf{A})$ for every $\mathbf{t} \in \mathbb{R}^3$.
- Rotation invariance: $\epsilon(\mathbf{X}, \mathbf{L}, \mathbf{A}) = \epsilon(\mathbf{X}, \mathbf{R}(\mathbf{L}), \mathbf{A})$ for every rotation matrix $\mathbf{R} \in O(3)$.
- Periodic cell choice invariance: $\epsilon(\mathbf{X}, \mathbf{L}, \mathbf{A}) = \epsilon(\mathbf{C}^{-1}\mathbf{X}, \mathbf{L}\mathbf{C}, \mathbf{A})$, where $\mathbf{C}$ triangular with $\det \mathbf{C} = 1$ and $\mathbf{C} \in \mathbb{Z}^{3 \times 3}$. See Fig. A1 for an example.
- Supercell invariance: $\epsilon(\mathbf{X}, \mathbf{L}, \mathbf{A}) = \epsilon\left(\bigoplus_{i=0}^{\det(\mathbf{C})} \mathbf{C}^{-1}(\mathbf{X} + \mathbf{k}_i \mathbf{1}_n^\top), \mathbf{L}\mathbf{C}, \bigoplus_{i=0}^{\det(\mathbf{C})} \mathbf{A}\right)$, where $\mathbf{C}$ is a 3×3 diagonal matrix with positive integers on the diagonal, $\mathbf{k}_i \in \mathbb{N}^3$ indexes the cell repetitions in the three lattice components, and $\bigoplus$ indicates concatenation.

Forces are instead equivariant to permutation and rotation, while being invariant to translation and periodic cell choice. Stress tensors are similarly invariant to permutation, translation, supercell choice, and periodic cell choice; while being equivariant to rotation (see Appendix A.8.1 for additional details).

A.3 Diffusion model background and notation

Diffusion models [41, 42, 64, 65] are a class of generative models that learn to revert a diffusion process. The diffusion process (also called the *forward* process) gradually corrupts an input sample $\mathbf{x}_0$ via transition kernels $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ ¹, defining a Markov chain $\mathbf{x}_0 \rightarrow \mathbf{x}_1 \rightarrow \dots \rightarrow \mathbf{x}_T$, where $T \in \mathbb{N}$ is the number of diffusion steps and $1 \leq t \leq T$. Here, we cover the typical case where the data is continuous-valued and the transition kernels are normal distributions. See Appendix A.5 for details on discrete diffusion models.

¹We follow the convention in machine learning literature that the functional forms of (conditional) probability density functions depend on the variables that appear as arguments. For example, $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ could be written as $q_{\mathbf{x}_t|\mathbf{x}_{t-1}}(\mathbf{x}_t|\mathbf{x}_{t-1})$ to make the dependence of the functional form on t explicit, but we avoid this to prevent clutter.

The transition kernels are of the general form $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(f(\mathbf{x}_{t-1}, t), \sigma_t^2 \mathbf{I})$, where $f(\mathbf{x}_{t-1}, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is affine in $\mathbf{x}_{t-1}$. This implies that the one-shot transition kernel $q(\mathbf{x}_t|\mathbf{x}_0)$ is also Gaussian, and for popular choices $f(\cdot, t)$ the mean and variance are known in closed form. This enables us to efficiently obtain a noisy sample $\mathbf{x}_t$ at an arbitrary time step t during training.

Diffusion models are optimized to approximate the score function of the noise distributions $q(\mathbf{x}_t|\mathbf{x}_0)$ for any $1 \leq t \leq T$:

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \sum_{t=1}^T \sigma_t^2 \mathbb{E}_{q(\mathbf{x}_0)} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0)\|_2^2], \quad (\text{A4})$$

where $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t) : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}^d$ is called the *score model*. It is standard practice [41, 42] to parameterize the model to predict the *noise* $\epsilon_t = -\sigma_t \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0)$ instead of the score, since the magnitude of $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is independent of the diffusion time step t .

The forward diffusion process is designed such that $q(\mathbf{x}_T|\mathbf{x}_0) \approx q(\mathbf{x}_T)$, where $q(\mathbf{x}_T)$ is a prior distribution that is easy to sample from (e.g., Gaussian).

In this work we leverage two popular diffusion processes for continuous data, i.e., the variance-exploding diffusion [41, 66] and the variance-preserving diffusion [42, 64] process, which we briefly explain in the following.

Variance-exploding diffusion

This is the diffusion process used in denoising score matching (DSM) [41]. We define a sequence of exponentially increasing standard deviations $\sigma_{\min} = \sigma_1, \dots, \sigma_T = \sigma_{\max}$ that define the transition kernels:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t, (\sigma_t^2 - \sigma_{t-1}^2) \mathbf{I}), \quad q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2 \mathbf{I}). \quad (\text{A5})$$

We can generate a sample using the learned model via annealed Langevin dynamics [41, 66] or ancestral sampling from the graphical model $\prod_{t=1}^T p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ [43]:

$$\mathbf{x}_{t-1} = \mathbf{x}_t + (\sigma_t^2 - \sigma_{t-1}^2) \mathbf{s}_{\boldsymbol{\theta}^*}(\mathbf{x}_t, t) + \mathbf{z} \sqrt{\sigma_t^2 - \sigma_{t-1}^2}, \quad (\text{A6})$$

where $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \sigma_T^2 \mathbf{I})$, and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is standard Gaussian noise. In Appendix A.6 we explain how we leverage variance-exploding diffusion in the diffusion process of the fractional coordinates.

Variance-preserving diffusion

This is the diffusion process used to train denoising diffusion probabilistic models (DDPMs) [42, 64]. In variance-preserving diffusion we define a sequence of positive noise scales $0 < \beta_1, \beta_2, \dots, \beta_T < 1$ to obtain transition kernels of the form

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (\text{A7})$$

where $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$. Sampling from a model trained to revert the variance-preserving diffusion process also works via *ancestral sampling* from the graphical model $\prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} (\mathbf{x}_t + \beta_t \mathbf{s}_{\theta^*}(\mathbf{x}_t, t)) + \sqrt{\beta_t} \mathbf{z}, \quad (\text{A8})$$

starting from $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is standard Gaussian noise. See Appendix A.7 for details about how we leverage variance-preserving diffusion in the diffusion process of the lattice.

A.4 Joint diffusion process

To apply the construction of a diffusion process described in Appendix A.3 to crystal structures described in Appendix A.1, we define the forward process through a Markov chain $\mathbf{M}_0 \rightarrow \mathbf{M}_1 \rightarrow \dots \rightarrow \mathbf{M}_T$ via a transition kernel that diffuses the atom coordinates, atom types, and the lattice independently as follows:

$$\begin{aligned} q(\mathbf{A}_{t+1}, \mathbf{X}_{t+1}, \mathbf{L}_{t+1} | \mathbf{A}_t, \mathbf{X}_t, \mathbf{L}_t) \\ = q(\mathbf{A}_{t+1} | \mathbf{A}_t) q(\mathbf{X}_{t+1} | \mathbf{X}_t) q(\mathbf{L}_{t+1} | \mathbf{L}_t) \quad (t = 0, 1, \dots, T-1). \end{aligned} \quad (\text{A9})$$

In addition, the noise distributions of atom species $\mathbf{A}$ and the fractional coordinates $\mathbf{X}$ factorize into the diffusion of the individual atoms:

$$q(\mathbf{A}_{t+1} | \mathbf{A}_t) = \prod_{i=1}^n q(a_{t+1}^i | a_t^i), \quad q(\mathbf{X}_{t+1} | \mathbf{X}_t) = \prod_{i=1}^n q(\mathbf{x}_{t+1}^i | \mathbf{x}_t^i).$$

Note that the factorization of the forward diffusion process does not imply that the reverse diffusion process factorizes in the same way. Details of the atom type diffusion, coordinate diffusion, and lattice diffusion are described in Appendix A.5, Appendix A.6, Appendix A.7, respectively. The architecture of the score network $s_{\theta}(\mathbf{M}_t, t)$ is described in Appendix A.8. The combined objective function is presented in Appendix A.9.

A.5 Atom type diffusion

For the diffusion of the (discrete) atom species $\mathbf{A}$, we use the discrete denoising diffusion probabilistic model (D3PM) approach [65], which is a generalization of DDPMs to discrete data problems. As in DDPM, the forward diffusion process is a Markov process that gradually corrupts an input sample a_0 , which is a scalar discrete random variable with K categories (e.g., atomic species):

$$q(a_{1:T} | a_0) = \prod_{t=1}^T q(a_t | a_{t-1}), \quad (\text{A10})$$

where $a_0 \sim q(a_0)$ is an atomic species sampled from the data distribution and $a_T \sim q(a_T)$, where $q(a_T)$ is a prior distribution that is easy to sample from.

Denoting the one-hot representation of a as a row vector $\mathbf{a}$, we can express the transitions as:

$$q(\mathbf{a}_t | \mathbf{a}_{t-1}) = \text{Cat}(\mathbf{a}_t; \mathbf{p} = \mathbf{a}_{t-1} \mathbf{Q}_t), \quad (\text{A11})$$

where $[\mathbf{Q}_t]_{ij} = q(a_t = j | a_{t-1} = i)$ is the Markov transition matrix at time step t . $\text{Cat}(\mathbf{a}; \mathbf{p})$ is a categorical distribution over one-hot vectors whose probabilities are given by the row vector $\mathbf{p}$. Similar to DDPM, D3PM assumes that the forward diffusion factorizes over all discrete variables of a data point, i.e., all atomic species are diffused independently with the same transition matrices $\mathbf{Q}_t$. Hence, we only consider individual one-hot vectors in this section. D3PMs are trained by optimizing a variational lower bound:

$$\begin{aligned} L_{\text{vb}} = \mathbb{E}_{q(\mathbf{a}_0)} & \left[-\mathbb{E}_{q(\mathbf{a}_1 | \mathbf{a}_0)} \log p_{\theta}(\mathbf{a}_0 | \mathbf{a}_1, 1) + D_{\text{KL}} [q(\mathbf{a}_T | \mathbf{a}_0) \parallel q(\mathbf{a}_T)] \right. \\ & \left. + \sum_{t=2}^T \mathbb{E}_{q(\mathbf{a}_t | \mathbf{a}_0)} D_{\text{KL}} [q(\mathbf{a}_{t-1} | \mathbf{a}_t, \mathbf{a}_0) \parallel p_{\theta}(\mathbf{a}_{t-1} | \mathbf{a}_t)] \right]. \end{aligned} \quad (\text{A12})$$

Moreover, Austin et al. [65] propose an additional cross-entropy loss on the model's prediction $p_{\theta}(\mathbf{a}_0 | \mathbf{a}_t, t)$:

$$L_{\text{CE}} = -\mathbb{E}_{q(\mathbf{a}_0)} \left[\sum_{t=2}^T \mathbb{E}_{q(\mathbf{a}_t | \mathbf{a}_0)} \log p_{\theta}(\mathbf{a}_0 | \mathbf{a}_t, t) \right],$$

so that the overall loss becomes

$$L = L_{\text{vb}} + \lambda_{\text{CE}} L_{\text{CE}}. \quad (\text{A13})$$

Three important characteristics of DDPM and DSM are that (1) given $\mathbf{x}_0$ we can sample noisy samples $\mathbf{x}_t$ for arbitrary t in constant time; (2) after sufficiently many diffusion steps, $\mathbf{x}_T$ follows a prior distribution that is easy to sample from; and (3) the posterior $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ in Eq. (A12) is tractable and can be computed efficiently. D3PM also has these properties, as we briefly outline in the following:

- (1) Fast sampling of $\mathbf{a}_t \sim q(\mathbf{a}_t | \mathbf{a}_0)$. Since the forward diffusion in D3PM is governed by discrete transition matrices $\{\mathbf{Q}_t\}_{t=1}^T$, we can write

$$q(\mathbf{a}_t | \mathbf{a}_0) = \text{Cat}(\mathbf{a}_t; \mathbf{p} = \mathbf{a}_{t-1} \bar{\mathbf{Q}}_t), \quad \text{where } \bar{\mathbf{Q}}_t = \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t. \quad (\text{A14})$$

The cumulative transition matrices $\bar{\mathbf{Q}}_t$ can be pre-computed and for many diffusion processes even have a closed form.

- (2) Tractable prior distribution. Two of the proposed diffusion processes are the absorbing (which we employ in MatterGen) and uniform diffusion processes. Both

gradually diffuse the data towards a limit distribution, which are the one-hot distribution on the absorbing state and the uniform distribution over all categories, respectively. For more details, we refer to Appendix A of Austin et al. [65].

- (3) Tractable posterior $q(\mathbf{a}_{t-1}|\mathbf{a}_t, \mathbf{a}_0)$. Using Bayes' rule and exploiting the Markov property $q(\mathbf{a}_t|\mathbf{a}_{t-1}, \mathbf{a}_0) = q(\mathbf{a}_t|\mathbf{a}_{t-1})$, we can write

$$q(\mathbf{a}_{t-1}|\mathbf{a}_t, \mathbf{a}_0) = \frac{q(\mathbf{a}_t|\mathbf{a}_{t-1})q(\mathbf{a}_{t-1}|\mathbf{a}_0)}{q(\mathbf{a}_t|\mathbf{a}_0)}. \quad (\text{A15})$$

All terms in Eq. (A15) can be computed efficiently in closed form given the forward diffusion process.

Reverse sampling process.

We generate a sample $\mathbf{a}_0$ by first sampling $\mathbf{a}_T$ and then gradually updating it to obtain $p_\theta(\mathbf{a}_{0:T}) = q(\mathbf{a}_T) \prod_{t=1}^T p_\theta(\mathbf{a}_{t-1}|\mathbf{a}_t)$. Austin et al. [65] propose to parameterize $p_\theta(\mathbf{a}_{t-1}|\mathbf{a}_t)$ by predicting a distribution over $\mathbf{a}_0$ and then marginalizing it out:

$$p_\theta(\mathbf{a}_{t-1}|\mathbf{a}_t) \propto \sum_{\mathbf{a}_0} q(\mathbf{a}_{t-1}, \mathbf{a}_t|\mathbf{a}_0) p_\theta(\mathbf{a}_0|\mathbf{a}_t, t), \quad (\text{A16})$$

where we can use our tractable posterior computation again. Since we have a discrete state space, marginalizing out $\mathbf{a}_0$ by explicit summation has complexity $\mathcal{O}(K)$. In the case of atomic species we have $K \simeq 100$; thus, this is relatively cheap. This parameterization has the advantage that potential sparsity in the diffusion process is efficiently enforced by using $q(\mathbf{a}_{t-1}, \mathbf{a}_t|\mathbf{a}_0)$ without having to be learned by the model.

Forward diffusion process.

As the specific flavor of D3PM forward diffusion we employ the masked diffusion process, which has shown best performance in the original study [65] as well as our initial experiments. Following Austin et al. [65], we introduce an extra atom species [MASK] at index $K - 1$, which is the absorbing or masked state. At each timestep t , the transition matrices have the particularly simple form

$$[Q_t^{\text{absorbing}}]_{ij} = \begin{cases} 1 & \text{if } i = j = m, \\ 1 - \beta_t & \text{if } i = j \neq m, \\ \beta_t & \text{if } j = m \neq i, \\ 0 & \text{if } m \neq i \neq j \neq m, \end{cases} \quad (\text{A17})$$

where m corresponds to the absorbing state. Intuitively, each species has probability $1 - \beta_t$ of staying unchanged, and probability β_t of transitioning to the absorbing state. Once a species is absorbed, it can never leave that state, and there are no transitions between different non-masked atomic species. Thus, the limit distribution of this diffusion process is a point mass on the absorbing state.